

Dictionaries in Python

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Introduction to Python Dictionaries

- Dictionaries is a collection of key-value pairs
- Ordered* (since Python 3.7)
- Changeable
- Do not allow duplicates

*It means that the items in a dictionary **maintain the order** in which they were inserted

Creating a Dictionary

- ▶ Syntax:

```
thisdict = {  
    "brand": "Mahindra",  
    "model": "New",  
    "year": 2015  
}
```

- ▶ Using type()

- ▶ Using len()

- ▶ Using dict constructor

```
thisdict = dict(name = "Asif", age = 36, country  
= "India")
```

Accessing Dictionary Items

- ▶ Using key name:

```
x = thisdict["model"]
```

- ▶ Using get() method:

```
x = thisdict.get("model")
```

- ▶ Can also store lists and sub-dictionary or number

```
myDict = {  
    "profile": "A Coder",  
    "marks": [1, 2, 5],  
    "anotherdict": {'work': 'Player'},  
    1: 205  
}
```

Modifying Dictionaries

- ▶ Adding items:

```
thisdict["color"] = "red"
```

- ▶ Changing values:

```
thisdict["year"] = 2020
```

- ▶ Updating a dictionary

```
updateDict = {  
    "hafiz": "Brother",  
    "sobia": "Friend",  
    "Suhail": "Friend",  
    "basit": "Cousin"  
}  
thisdict.update(updateDict)
```

Dictionary Methods

- ▶ `keys()`: Returns all keys
`mydict.keys()`
- ▶ `values()`: Returns all values
`mydict.values()`
- ▶ `items()`: Returns key-value pairs
`mydict.items()`
- ▶ `update()`: Updates with new values
`mydict.update({"color": "red"})`
- ▶ `copy()`: Copies dictionary
`mydict = thisdict.copy()`
`print(mydict)`

Removing Items

- ▶ Using pop():

```
thisdict.pop("model")
```

- ▶ Using del:

```
del thisdict["model"]
```

- ▶ Using clear():

```
thisdict.clear()
```

- ▶ Using popitem(): Removes last inserted item

```
thisdict.popitem()
```

Nested Dictionaries

- ▶ `myfamily = {`
- ▶ `"child1": {"name": "Amir", "year": 2008},`
- ▶ `"child2": {"name": "Toyiba", "year": 2007},`
- ▶ `"child3": {"name": "Hafsa", "year": 2012}`
- ▶ `}`

Practice set

- ▶ Write a program to create a dictionary of computer device names with values as their definitions
- ▶ Write a program to create a dictionary that stores the names of favourite fruits of your 3 friends and print the dictionary

Summary

- ▶ Dictionaries are versatile in Python
- ▶ Useful for storing related data efficiently
- ▶ Essential for Python programming